Optimal Allocation of Observations across Run-In, Treatment, and Common Close Phases in Longitudinal Clinical Trials

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Abstract

Background. Trial designers planning a longitudinal study with a fixed total observation budget face an allocation question: how should observations be distributed across run-in (J_0), treatment (J_1), and common close (J_2) phases to minimize the variance of the treatment-effect estimator? Existing guidance is heuristic; the optimum has not been characterized for the three-phase run-in setting.

Methods. We formulate the allocation question as a discrete optimization problem over the Woodbury-based variance formula derived in Paper 1 of this compendium and solve it by exhaustive enumeration of all feasible phase splits. The behavior of the optimum is described as a function of the signal-to-noise ratio σ_b/σ. Extensions cover replicated endpoint observations (closely spaced measurements at the end of treatment that average out measurement error on the final assessment, analogous to run-in averaging applied at the post-treatment end) and pattern-mixture handling of monotone dropout following the Frost-Kenward-Fox (2008) framework.

Results. Under a fixed total budget the optimal design places zero observations in the run-in phase in every configuration examined; the budget is split between the treatment and common close phases. This reconciles with the run-in efficiency gains of Paper 1: run-in observations help when they are additional visits appended to a fixed treatment phase, but when visits are fungible the run-in phase is never chosen, because it introduces the pre-randomization slope parameter without contributing to the treatment contrast. The common close phase enters the optimal design once σ_b/σ exceeds a budget-dependent crossover, which falls from 0.81 at $J = 3$ to 0.30 at $J = 8$; at high ratios roughly half the budget goes to the common close. Replicating the final endpoint reduces the treatment-effect variance by up to 31% at $\sigma_b/\sigma = 0.5$ (five replicates) but by at most 5% at $\sigma_b/\sigma = 2$.

Conclusions. The numerical allocation results give trial designers a simple decision procedure driven by the signal-to-noise ratio and the visit budget. The procedure extends to replicated-endpoint and dropout-adjusted designs within the same variance framework.

Contents

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1 Introduction

[1] **Additional run-in and common close observations both improve efficiency but with diminishing or J_1-dependent returns.**

- Run-in observations appended to a fixed treatment phase improve efficiency with diminishing returns.
- Most of the attainable gain comes from the first two run-in observations; the marginal value falls quickly thereafter.
- Common close gains depend on the number of treatment observations J_1.

The relative efficiency surface in Paper 1 (Figure 1) demonstrates diminishing returns from additional run-in observations appended to a fixed treatment phase: most of the attainable variance reduction is recovered by the first two run-in observations, and the marginal value falls quickly thereafter (Section 3 examines the marginal pattern, including a penalty specific to a single run-in observation). Similarly, common close observations improve efficiency but at a rate that depends on the number of treatment observations J_1.

[2] **The efficiency results motivate an allocation question left open by Paper 1.**

- Trials operate under constraints on total visits or total duration.
- Paper 1 characterizes efficiency but not the optimal phase split.
- The question is how to distribute observations across the three phases.

These observations raise a design question that Paper 1 does not address: given operational constraints on the total number of visits or total trial duration, what is the optimal distribution of observations across the three phases?

[3] **Prior work addresses related design questions but not optimal allocation across run-in, treatment, and common close phases.**

- Existing optima target time-point positioning or general design guidelines, not the run-in setting.
- Multiple pre-treatment measurements are known to benefit ANCOVA-analyzed designs.
- A substantial optimal-design literature treats measurement schedules for mixed models through the Fisher information matrix, but does not address a phase structure in which one phase carries its own nuisance parameter.
- Available software computes slope-comparison power but does not optimize the phase allocation.

[1] calculated optimal time-point positions for linearly divergent treatment effects under compound symmetry, but did not consider the run-in setting.[2] derived sample sizes for trends across repeated measurements under missing data.[3] showed that having more than one pre-treatment measurement is beneficial for designs analyzed via ANCOVA.[4] provided sample size formulas for repeated measurement designs.[5] developed general guidelines for the design of clinical trials with longitudinal outcomes.

The question posed here is a design-optimization problem and belongs to a well-developed tradition. The general theory of optimum experimental design, in which a design is chosen to optimize a functional of the Fisher information matrix, is set out by[6] and[7].[8] extended the theory to random-effects regression models, the model class used here, giving an algorithm that maximizes the determinant of the population information matrix. Closest in setting is[9], who constructed D-optimal cohort designs for linear mixed-effects models with a cost function for repeated measurements, and found that concentrating measurements at optimally chosen time points in a single cohort is efficient. Three features distinguish the present problem from that literature. First, the design space is not a set of time points to be placed freely but a partition of a fixed count of equally spaced visits into three phases with a prescribed ordering. Second, the run-in phase introduces its own nuisance parameter, the pre-randomization slope δ, so enlarging that phase both adds information and consumes a degree of freedom; the optimal-design literature usually holds the parameter vector fixed as the design varies. Third, the criterion here is the variance of a single *scalar contrast, a c-optimality criterion, rather than the D-optimality that most of this literature adopts. The results below are accordingly numerical rather than an application of the general equivalence theory.*

The slopepower command[10] and the longpower R package[11] implement power calculations for slope-comparison trials but do not optimize the allocation of observations across phases.

2 Problem formulation

2.1 Fixed total visits

[4] **The fixed-visit problem minimizes the treatment-effect variance over the phase split under a total-count constraint.**

- The budget J excludes baseline and must be split as $J_0 + J_1 + J_2 = J$.
- Feasibility requires $J_0 \geq 0$, $J_1 \geq 1$, and $J_2 \geq 0$.
- The objective is the $\gamma\gamma$ entry of $(X'\Sigma^{-1}X)^{-1}$.

Given J total observations (excluding baseline), choose (J_0^*, J_1^*, J_2^*) to minimize $\mathrm{Var}_1(\hat{\gamma})$ subject to $J_0 + J_1 + J_2 = J$, $J_0 \geq 0$, $J_1 \geq 1$, $J_2 \geq 0$.

The objective function is $V(J_0, J_1, J_2) = [(X'\Sigma^{-1}X)^{-1}]_{\gamma\gamma}$ computed via the Woodbury identity from Paper 1.

Throughout, the phase counts (J_0, J_1, J_2) correspond to the arguments (r, k, f) of the `runinpower` package routines used in the code chunks and tables.

2.2 Fixed total duration

[5] **The fixed-duration problem instead constrains total trial length and may also optimize observation timing.**

- Duration is fixed via $T = (J_0 + J_1 + J_2) \cdot t$, so the count constraint becomes $J_0 + J_1 + J_2 = T/t$.
- The variance $\mathrm{Var}_1(\hat{\gamma})$ is minimized subject to this constraint.
- A general version permits non-equally-spaced observations and optimizes over the observation times.

Alternatively, fix the trial duration $T = (J_0 + J_1 + J_2) \cdot t$ and minimize $\mathrm{Var}_1(\hat{\gamma})$ subject to $J_0 + J_1 + J_2 = T/t$, or more generally allow non-equally-spaced observations within each phase and minimize over the observation times as well.

3 Diminishing returns from run-in observations

3.1 Marginal value of the J_0-th run-in observation

Define the marginal variance reduction from adding the J_0-th run-in observation:

$$\Delta V_{J_0} = V(J_0 - 1, J_1, J_2) - V(J_0, J_1, J_2) \tag{1}$$

Table 1: Marginal variance reduction from the J_0-th run-in observation ($J_1 = 2$, $J_2 = 0$, $\sigma^2 = 1$, $\sigma_b^2 = 1$).

J_0	ΔV_{J_0}
1	0.1765
2	0.6667
3	0.5124
4	0.3176
5	0.2008
6	0.1342
7	0.0945
8	0.0696

Table 2: Boundary of the $J_0 = 1$ anomaly ($J_1 = 2$, $J_2 = 0$, $\sigma = t = 1$). The reversal $\Delta V_1 < \Delta V_2$ holds only between the two computed boundary ratios 0.53 and 1.51.

σ_b/σ	ΔV_1	ΔV_2	$\Delta V_1 < \Delta V_2$
0.32	0.1633	0.0080	no
0.53	0.0454	0.0418	no
0.75	0.0033	0.2653	yes
1.00	0.1765	0.6667	yes
1.25	0.6649	1.1082	yes
1.51	1.5407	1.5352	no
2.00	4.2000	2.1718	no

3.2 The $J_0 = 1$ anomaly

[6] **In an intermediate signal-to-noise band the first run-in observation is less valuable than the second because it introduces a new model parameter.**

- At $J_1 = 2$, $J_2 = 0$, $\sigma = t = 1$, $\Delta V_1 < \Delta V_2$ holds for σ_b/σ between approximately 0.53 and 1.51, and fails outside this band.
- Adding one run-in observation expands the model from two to three parameters, consuming a degree of freedom.
- The second run-in observation adds no parameter and delivers pure information gain.

A non-monotonicity is visible in the table: $\Delta V_1 < \Delta V_2$ at the displayed parameter setting ($\sigma_b/\sigma = 1$). The first run-in observation is less valuable than the second. This occurs because adding a single run-in observation forces the model from 2 parameters (β, γ) to 3 parameters (δ, β, γ), consuming a degree of freedom. The second run-in observation does not increase the parameter count and provides pure information gain. The reversal is not universal: a numerical scan (Table 2) shows that, for $J_1 = 2$, $J_2 = 0$, and $\sigma = t = 1$, the inequality $\Delta V_1 < \Delta V_2$ holds only for σ_b/σ in an intermediate band, approximately $(0.53, 1.51)$; at lower and higher ratios the first run-in observation is the more valuable one.

[7] **The single-run-in penalty parallels a known ANCOVA result and**

yields a regime-specific design rule.

- @frison1992 found a single pre-treatment ANCOVA covariate can reduce power at low pre-post correlation.
- The same degree-of-freedom cost applies in the run-in setting.
- Within the intermediate band of Table reftab:anomaly-boundary, designers who add run-in observations should plan at least two rather than exactly one.

This effect is loosely analogous to[3]'s finding that a single pre-treatment measurement used as an ANCOVA covariate can reduce power when the pre-post correlation is low. The practical implication is regime-specific: ***within the intermediate signal-to-noise band of Table 2, a trial designer who adds run-in observations should plan at least two, not one.*** *Outside that band the first run-in observation carries the larger marginal value and no special caution applies.*

3.3 Correlation structure governs diminishing returns

The correlation between the J_0-th run-in change score and the treatment-period change scores determines the information gain. Under compound symmetry:

$$\mathrm{Corr}(C_{-J_0}, C_l) = \frac{\sigma_b^2 t^2}{2\sigma^2 + \sigma_b^2 t^2} \tag{2}$$

[8] **Under compound symmetry diminishing returns stem from the information matrix, whereas under AR(1) declining correlation accelerates them.**

- The run-in change-score correlation is constant in J_0 under compound symmetry.
- Diminishing returns for $J_0 \geq 2$ therefore arise from the information matrix structure.
- Under AR(1) the correlation falls with temporal distance, yielding faster diminishing returns from distant run-in observations.

which is constant in J_0. The diminishing returns (for $J_0 \geq 2$) arise from the information matrix structure, not from declining correlation. Under AR(1)[12], the correlation does decline with temporal distance, producing faster diminishing returns from distant run-in observations.

4 Optimal allocation for fixed J

[9] **Under a fixed total budget the optimum never allocates observations to the run-in phase.**

- In every cell of Table reftab:optimal-grid, $J_0^* = 0$; the budget splits between treatment and common close.
- The run-in phase adds the pre-randomization slope parameter δ without contributing to the treatment contrast, so a run-in visit is dominated by a treat-

Table 3: Optimal allocation (J_0^*, J_1^*, J_2^*) for fixed total observations J, by signal-to-noise ratio.

J	Low ($\sigma_b/\sigma = 0.32$)	Med ($\sigma_b/\sigma = 1.0$)	High ($\sigma_b/\sigma = 2.24$)	AD ($\sigma_b/\sigma = 1.70$)
3	(0,3,0)	(0,2,1)	(0,2,1)	(0,2,1)
4	(0,4,0)	(0,2,2)	(0,2,2)	(0,2,2)
5	(0,5,0)	(0,3,2)	(0,3,2)	(0,3,2)
6	(0,6,0)	(0,3,3)	(0,3,3)	(0,3,3)
7	(0,5,2)	(0,4,3)	(0,4,3)	(0,4,3)
8	(0,5,3)	(0,4,4)	(0,4,4)	(0,4,4)

ment or close visit whenever visits are fungible.

- Paper 1's run-in gains are not contradicted: they arise when run-in visits are additional, appended to a fixed treatment phase, not reallocated from it.

The dominant feature of Table 3 is that the optimal run-in count is zero in every cell: across all budgets $J = 3, \dots, 8$ and all four signal-to-noise settings, including the Alzheimer's neuroimaging regime, the optimizer splits the budget between the treatment and common close phases and never assigns a visit to the run-in phase. The mechanism is a degree-of-freedom cost. Any positive run-in count forces the model to estimate the pre-randomization slope δ in addition to (β, γ), and run-in observations inform the treatment contrast only indirectly, through the random slope; a visit placed in the treatment or common close phase buys the same slope information without the extra parameter. This finding does not contradict the efficiency gains of Paper 1. There, run-in observations are additional visits appended to a design whose treatment phase is held fixed, and the comparison is against the shorter design without them. Here visits are fungible under a fixed total budget, and a run-in visit always has a better use. The two framings answer different design questions, and the distinction is the central practical message of this paper.

5 When is the common close phase beneficial?

[10] **The common close phase aids estimation only indirectly, through the random slope, so the meaningful question is the fixed-budget trade-off.**

- Common close observations inform the random slope b_i; they do not contribute to the treatment comparison directly.
- Added to a fixed design, a common close observation reduced $\mathrm{Var}(\hat{\gamma})$ in every configuration examined ($J_1 = 1, \dots, 4$, σ_b/σ down to 10^{-3}), so no benefit threshold exists for free observations.
- The design-relevant question is whether a budgeted visit is better spent on the common close than on the treatment phase.

The common close phase contributes information about the random slope b_i without contributing to the treatment comparison directly. When a common close observation is added to an otherwise fixed design, it reduced $\mathrm{Var}(\hat{\gamma})$ in every configuration we examined ($J_1 = 1, \dots, 4$ with two run-in observations, and σ_b/σ down to 10^{-3}): a free observation is always at least weakly helpful. The design-relevant question is therefore not whether the common close helps in isolation, but whether a visit taken from a fixed budget is better spent on the common close than on the treatment phase. That trade-off is quantified next.

Table 4: Signal-to-noise ratio above which reallocating the last budgeted visit from the treatment phase to the common close, $(0, J-1, 1)$ versus $(0, J, 0)$, reduces $\mathrm{Var}(\hat{\gamma})$.

J	Crossover σ_b/σ
3	0.806
4	0.592
5	0.471
6	0.392
7	0.336
8	0.295

5.1 Crossover as a function of the budget J

[11] **The crossover ratio falls as the budget grows, and above it the close phase absorbs an increasing share of the budget.**

- The crossover decreases from 0.81 at $J = 3$ to 0.30 at $J = 8$.
- Above the crossover the full optimizer allocates a growing share to the close, reaching an even split $(0, 4, 4)$ at $J = 8$ at every examined ratio from 0.6 upward.
- Even at $\sigma_b/\sigma = 0.32$ the optimum at $J = 8$ is $(0, 5, 3)$, so the close enters at low ratios once the budget is large.

Table 4 shows the crossover ratio at which the reallocation of a single budgeted visit from the treatment phase to the common close becomes favorable. The crossover falls monotonically with the budget, from $\sigma_b/\sigma \approx 0.81$ at $J = 3$ to ≈ 0.30 at $J = 8$: larger trials should commit visits to the common close at progressively lower signal-to-noise ratios. Above the crossover the full optimizer (Table 3) assigns an increasing share of the budget to the close phase; at $J = 8$ the optimum is an even split $(0, 4, 4)$ at every ratio examined from 0.6 upward, and even at $\sigma_b/\sigma = 0.32$ the optimum is $(0, 5, 3)$. These values are numerical results from exhaustive enumeration; an analytical characterization of the crossover remains open.

6 Non-equally-spaced designs

6.1 General time points

The display in this section comes from a Mathematica notebook derivation for a *different* model than the one used elsewhere in this paper: a raw-outcome random-slope-only model with centered observation times ($\sum t_i = 0$) and independent residuals $R = \sigma^2 I$, rather than the change-score model with shared baseline error $R = \sigma^2(I + \mathbf{1}\mathbf{1}')$ that underlies all preceding sections. Under that raw-outcome model, with the normalization $\sum t_i^2 = 1$, the notebook gives for two time points t_1, t_2:

$$\mathrm{Var}(\hat{\gamma}) = \frac{3\sigma^2}{1 - t_1 t_2} + 2\sigma_b^2 \tag{3}$$

This expression has not been re-derived or numerically verified within the change-score framework of this paper, and reconciling the two formulations remains open.

[12] **Maximally separated time points minimize the slope-estimation variance, consistent with the classical extremes result.**

- Within the raw-outcome model, the variance is minimized when $t_1 t_2$ is most negative.
- This occurs when the two time points are maximally separated.
- It is consistent with the known optimality of observations at the extremes of follow-up for slope estimation.

Within the raw-outcome model above, the expression is minimized when $t_1 t_2$ is minimized (most negative), i.e., when the two time points are maximally separated. This is consistent with the known result that observations at the extremes of the follow-up period are optimal for slope estimation. Whether and how this carries over to the change-score model of the preceding sections has not been established here.

6.2 Optimization over observation times within phases

7 Replicated endpoint observations

7.1 Motivation

[13] **The averaging logic that motivates run-in can be applied symmetrically at the end of treatment to reduce endpoint measurement error.**

- Run-in averages pre-randomization observations to cut the measurement-error contribution to the baseline slope.
- Replicated measures raise power with diminishing returns past 20-50 replications.
- Closely spaced replicates of the final visit average out endpoint measurement error under the same treatment condition.

The run-in mechanism averages multiple pre-randomization observations to reduce the contribution of measurement error σ^2 to the estimated baseline slope[3,13].[14] showed that replicated measures increase power but with diminishing returns past 20-50 replications. The same logic applies at the end of the treatment period: if the final scheduled visit is replaced by (or augmented with) two or three closely-spaced measurements, all under the same treatment condition, the measurement error on the endpoint assessment is reduced by averaging.

[14] **Replicating an endpoint or baseline visit scales down its residual variance while leaving the random-slope contribution intact.**

- Replicating the last visit n_{rep} times over a short interval gives effective residual variance $\sigma^2/n_{\mathrm{rep}}$.
- The random slope contribution $\sigma_b^2 t^2$ is essentially unchanged.
- Replicating the baseline visit similarly reduces σ^2 in the change-score covariance.

Concretely, if the last visit at time $J_1 t$ is replicated n_{rep} times over a short interval (days to weeks, negligible relative to t), the effective residual variance for the averaged endpoint becomes σ^2/n_{rep} while the random slope contribution $\sigma_b^2 t^2$ is essentially unchanged.

Table 5: Fractional variance reduction from replicating the final endpoint ($J_0 = 0$, $J_1 = 2$, $J_2 = 0$).

n_{rep}	$\sigma_b/\sigma = 0.5$	$\sigma_b/\sigma = 1$	$\sigma_b/\sigma = 2$	$\sigma_b/\sigma = 5$
1	0.000	0.000	0.000	0.000
2	0.182	0.091	0.030	0.005
3	0.250	0.125	0.042	0.007
5	0.308	0.154	0.051	0.009

Similarly, the baseline visit at $j = 0$ can be replicated to reduce σ^2 in the change-score covariance.

[15] **Replicated endpoints differ fundamentally from common close observations in timing and the information they provide.**

- Common close observations occur at new time points after treatment stops.
- They inform the off-treatment slope.
- Replicated endpoints sit at the same time point as the last treatment visit and only reduce measurement error.

This is distinct from the common close design (Section 5): common close observations are at new *time points after treatment stops, providing information about the off-treatment slope. Replicated endpoints are at the* same *time point as the last treatment visit, providing only measurement-error reduction.*

7.2 Variance formula with replication

[16] **Replication enters only through modified entries of R, leaving the Woodbury computation otherwise unchanged.**

- The diagonal entry for a replicated visit uses σ^2/n_{rep} in place of σ^2.
- Off-diagonal entries reflecting shared baseline error use $\sigma^2/n_{\text{rep,pre}}$.
- The remaining Woodbury computation proceeds identically.

The residual covariance matrix R is modified so that the diagonal element corresponding to a replicated visit uses σ^2/n_{rep} instead of σ^2, and the off-diagonal elements reflecting the shared baseline error use $\sigma^2/n_{rep,pre}$. The Woodbury computation proceeds identically.

[17] **Endpoint replication helps most when measurement error dominates and little when slope variability does.**

- The benefit is largest at small σ_b/σ.
- At large $\sigma_b/\sigma > 2$ the gain is modest.
- Variance is then dominated by $\sigma_b^2 t^2$, which replication cannot reduce.

The table confirms that endpoint replication is most beneficial when measurement error dominates (σ_b/σ small). When between-subject slope variability is large ($\sigma_b/\sigma > 2$), the gain from replication is modest because the variance is dominated by $\sigma_b^2 t^2$, which replication cannot reduce.

7.3 Replication vs additional run-in observations

[18] **A fixed budget of extra measurements can be allocated either to run-in observations or to endpoint replicates.**

- Run-in observations sit at distinct time points separated by t.
- Endpoint replicates are closely spaced at one time point.
- Replicates provide measurement-error reduction only, not slope information.

A natural question is whether a fixed budget of additional measurements should be spent on run-in observations (at distinct time points, separated by t) or on endpoint replicates (closely spaced, measurement-error reduction only).

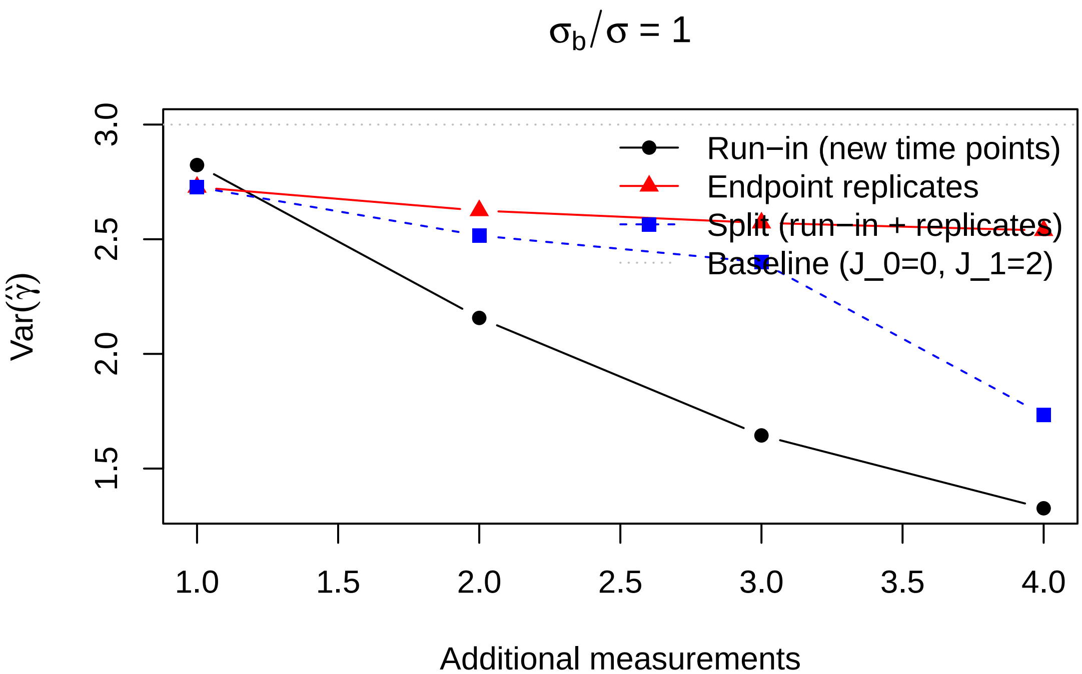

Figure 1: (#fig:rep_vs_runin)Variance as a function of additional measurements allocated either to run-in (new time points) or endpoint replication (same time point). Baseline: $J_0 = 0$, $J_1 = 2$, $\sigma^2 = 1$, $\sigma_b^2 = 1$.

[19] **Whether run-in or endpoint replication wins depends on the signal-to-noise ratio and on the number of extra measurements.**

- At $\sigma_b/\sigma = 1$, a single extra measurement is better spent on a replicate (variance 2.73 versus 2.82); run-in wins from the second extra measurement onward.
- Run-in observations supply information about the individual slope b_i, not only measurement-error reduction, but a lone run-in observation pays the single-run-in penalty of Section 3.
- At $\sigma_b/\sigma = 0.5$, replication won at every budget examined (one to four extra measurements) because measurement error dominates the variance.

At $\sigma_b/\sigma = 1$ (the setting of the figure), the comparison depends on how many extra measurements are available. With a single extra measurement, the endpoint replicate is the better use (variance 2.73 versus 2.82 for one run-in observation): the lone run-in observation pays the degree-of-freedom penalty of Section 3. From the second extra

Table 6: Per-pair variance with baseline and endpoint replication ($J_0 = 0$, $J_1 = 2$, $\sigma^2 = 1$, $\sigma_b^2 = 1$).

n_{pre}	$n_{\text{post}} = 1$	$n_{\text{post}} = 2$	$n_{\text{post}} = 3$
1	3.0000	2.7273	2.6250
2	2.7273	2.5000	2.4138
3	2.6250	2.4138	2.3333

measurement onward, run-in observations at new time points win (2.16 versus 2.63 at two extras, 1.64 versus 2.57 at three), because they provide information about the individual slope b_i, not just measurement-error reduction. When $\sigma_b/\sigma < 1$, endpoint replication is stronger still: at $\sigma_b/\sigma = 0.5$ it beat run-in at every budget we examined (one to four extra measurements), since measurement error is then the dominant source of variance.

7.4 Combined replication at both ends

[20] **Replicating both terminal visits yields symmetric measurement-error reduction.**

- Baseline replication reduces σ^2 at the start of follow-up.
- Endpoint replication reduces σ^2 at the end of treatment.
- Together they lower measurement error symmetrically at both ends.

Replicating both the baseline and endpoint visits provides symmetric measurement-error reduction.

8 Incorporating dropout

8.1 Pattern-mixture approach

[21] **Dropout is accommodated by a pattern-mixture model that combines stratum-specific sample sizes.**

- The approach follows @frost2008 Section 2.3 and related pattern-mixture work.
- Participants are stratified by their observed visit pattern.
- The overall requirement is a weighted average across strata.

Following[13] (Section 2.3), dropout is handled via a pattern-mixture model[15,16]. Participants are stratified by their observed visit pattern, and the overall variance is a weighted average:

$$N = \frac{1}{\sum_s p_s / N_s} \tag{4}$$

[22] **Strata are truncated designs under monotone dropout, and the formula is a conservative approximation that exact treatments improve upon.**

Table 7: Sample size adjusted for dropout via pattern-mixture ($J_0 = 2$, $J_1 = 2$, $J_2 = 1$, MIRIAD parameters).

Annual dropout	N (no dropout)	N (with dropout)
0.0	104	104
0.1	104	128
0.2	104	160
0.3	104	204

- Here p_s is the proportion with pattern s and N_s the size needed if all had pattern s.
- Under monotone dropout, the pattern after post-baseline visit j is the truncated design retaining the first j post-baseline observations; participants with no post-baseline visit contribute no slope-contrast information ($N_s = \infty$).
- @hu2021 derive exact formulas under monotone dropout, more rigorous and less conservative when dropout is informative about the random slope.

where p_s is the proportion with pattern s and N_s is the sample size that would be needed if all participants had pattern s. Under monotone dropout, the pattern for a participant who leaves after post-baseline visit j is the truncated design that retains the run-in and baseline visits plus the first j post-baseline observations; a participant with no post-baseline visit contributes no information about the treatment contrast, so that stratum enters with $N_s = \infty$. Dropout during the pre-randomization run-in is ignored here, since run-in dropouts are typically replaced before randomization. This approximation treats each stratum independently.[17] derive exact variance formulas under monotone dropout for the three-component random coefficient regression model ($\sigma^2, \sigma_a^2, \sigma_b^2$), accounting for the effect of dropout on the estimability of the random slope variance. Their approach is more rigorous than the pattern-mixture approximation used here, which is conservative when dropout is informative about the random slope.

9 Numerical examples

9.1 AD neuroimaging trial

[23] **The AD example evaluates optimal allocation under MIRIAD parameters across several budgets and dropout scenarios.**

- Parameters are $\sigma^2 = 0.3363$, $\sigma_b^2 = 0.9745$, and $t = 1$ year.
- Allocation is compared for $J = 4, 5, 6$ total observations.
- Both with-dropout and without-dropout cases are considered.

Using MIRIAD parameters ($\sigma^2 = 0.3363$, $\sigma_b^2 = 0.9745$, $t = 1$ year), compare the optimal allocation for $J = 4, 5, 6$ total observations with and without dropout.

10 Discussion

[24] **Under a fixed visit budget the optimum places nothing in the run-in phase, which reconciles rather than contradicts the run-in literature.**

Table 8: Top 10 allocations for $J = 5$ observations (MIRIAD parameters, $\Delta = 0.25$, 90% power).

(J_0, J_1, J_2)	$\mathrm{Var}_1(\hat{\gamma})$	N per group
(0,3,2)	0.291783	50
(0,2,3)	0.303230	51
(1,2,2)	0.453050	77
(2,3,0)	0.551570	93
(0,4,1)	0.560024	95
(1,3,1)	0.587888	99
(3,2,0)	0.593958	100
(2,2,1)	0.616148	104
(0,1,4)	0.701145	118
(1,1,3)	0.803845	136

- In every configuration examined the optimizer sets $J_0^* = 0$ and splits the budget between treatment and common close.
- Run-in visits pay for the pre-randomization slope parameter δ while informing the treatment contrast only indirectly, so a fungible visit is always better spent elsewhere.
- Paper 1's efficiency gains apply when run-in visits are additional; the two framings answer different design questions.

The central allocation result is negative and, we believe, useful: under a fixed total visit budget, the optimal design places no observations in the run-in phase in any configuration examined, including the Alzheimer's neuroimaging regime (Table 3). The budget splits between the treatment and common close phases. The mechanism is the degree-of-freedom cost identified in Section 3: any positive run-in count adds the pre-randomization slope parameter δ, while run-in observations inform the treatment contrast only indirectly. This does not contradict the efficiency gains reported for run-in designs in Paper 1 and the prior literature; those gains arise when run-in visits are additional measurements appended to a fixed treatment phase. When visits are fungible, as under a per-participant visit budget, the same visit buys more information in the treatment or common close phase. Trial designers should therefore ask which question their operational constraint poses: extra visits (run-in can help) or reallocated visits (run-in never helps in the settings examined here).

[25] **The signal-to-noise ratio and the budget jointly determine how much of the budget the common close should absorb.**

- Reallocating one visit from treatment to common close pays once σ_b/σ exceeds a crossover that falls from 0.81 at $J = 3$ to 0.30 at $J = 8$.
- Above the crossover the close phase absorbs a growing share, reaching an even split at $J = 8$ in the settings examined.
- Designers should estimate the slope-to-noise ratio from pilot data before fixing the phase split.

The common close allocation is governed jointly by the signal-to-noise ratio and the budget. A free common close observation always helped in the configurations we examined (Section 5), so the operative question is the budgeted trade-off: moving one visit from the treatment *phase to the common close reduces the variance once σ_b/σ exceeds a crossover that falls with the budget, from about 0.81 at $J = 3$ to 0.30 at $J = 8$ (Table 4). Above the crossover the optimizer assigns the close phase a growing share of the budget, up to an even split at $J = 8$ in the settings examined. In practice, high slope-to-noise outcomes such as neuroimaging endpoints sit well above the crossover at realistic budgets, so a substantial common close phase is warranted there, while low slope-to-noise outcomes justify a common close only at larger budgets. Estimating σ_b/σ from pilot data is the prerequisite design step.*

[26] **Two refinements, unequal spacing and endpoint replication, offer further efficiency without changing the phase counts.**

- In the raw-outcome special case of Section 6, spacing two observations toward the extremes of the window minimizes the slope-estimation variance; the general change-score case remains open.
- Replicating the endpoint or baseline visit reduces measurement error and is most valuable when measurement error dominates.
- At $\sigma_b/\sigma = 1$, a single extra measurement favors an endpoint replicate; run-in observations at new time points win from the second extra measurement onward.

Two refinements operate within a fixed phase allocation. First, when observation times within a phase are free, the two-point raw-outcome special case of Section 6 indicates that spacing observations toward the extremes of the window minimizes the slope-estimation variance; the corresponding optimization in the change-score model of this paper remains open. Second, replicating the baseline or endpoint visit reduces the measurement-error contribution to the change score without adding new time points; this is most valuable at low slope-to-noise ratios (up to a 31 percent variance reduction at $\sigma_b/\sigma = 0.5$ with five replicates) and offers little when slope variability dominates (at most 5 percent at $\sigma_b/\sigma = 2$). The comparison of Section 7 clarifies the choice between spending an additional measurement on a new run-in time point versus an endpoint replicate: at $\sigma_b/\sigma = 1$ the replicate wins when only one extra measurement is available, because a lone run-in observation pays the degree-of-freedom penalty, while run-in observations win from the second extra measurement onward by contributing slope information; below $\sigma_b/\sigma = 1$ replication won at every budget examined.

[27] **The allocation rules are derived under idealizing assumptions whose relaxation is the natural next step.**

- The optima assume equally spaced observations, a linear trajectory, and compound-symmetric residuals.
- Dropout is handled by a pattern-mixture approximation over monotone truncation strata, which the exact treatment of @hu2021 would refine.
- Under AR(1) residuals the diminishing returns to distant run-in observations are steeper, shifting the optimum toward fewer, closer run-in observations.

Several assumptions bound the scope of these recommendations. The allocation optima are computed under equally spaced observations, linear individual trajectories, and a compound-symmetric residual structure; the AR(1) analysis of Paper 2 implies that under decaying correlation the optimum shifts toward fewer and more closely spaced run-in observations, since distant ones lose informativeness. The allocation results themselves are numerical, *obtained by exhaustive enumeration of the feasible phase splits; an analytical characterization of the optimum, for example by signing the marginal value of a close-phase visit, remains open. Dropout is accommodated through the pattern-mixture approximation of Section 8, applied to monotone truncation strata, which the exact monotone-dropout treatment of*[17] *would refine. Extending the allocation optimization jointly over phase counts, within-phase timing, and the correlation structure remains the natural next step.*

11 Conflict of interest

[28] **The author reports no conflicts of interest.**

- No financial conflicts are present.
- No non-financial conflicts are present.
- No competing interests bear on this work.

The author declares no conflicts of interest.

12 Funding

[29] **The work received no specific funding.**

- No external grant supported the study.
- No internal grant supported the study.
- No sponsor influenced the analysis.

This work received no specific funding.

13 Data availability statement

[30] **The implementing package, tests, and shared bibliography are openly available in the project repository.**

- Allocation routines reside in allocation.R and replicated.R; design.R builds the design matrices they rely on.
- Validation tests and the four-paper bibliography are included.
- Specific paths appear in the Reproducibility section below.

The `runinpower` R package implementing the optimal-allocation routines (`allocation.R` and `replicated.R`, with the design matrices they rely on built in `design.R`), validation tests, and the shared bibliography across the four-paper compendium are openly available in the project repository. Specific paths are listed in the Reproducibility section below.

14 Reproducibility

[31] **The manuscript build relies on a small, fixed set of R packages.**

- Rendering used R 4.4.x.
- The in-package runinpower source and tinytest provide computation and testing.

- rmarkdown and bookdown drive the manuscript build.

This manuscript was rendered with R 4.4.x. Principal packages used are the in-package `runinpower` source, `tinytest` (testing harness), and `rmarkdown` plus `bookdown` for the manuscript build.

[32] **All results are deterministic.**

- Every table and figure is a deterministic evaluation of the variance formulas.
- No Monte Carlo simulation appears in this manuscript.
- No random seed is therefore required.

Random-number generator. *All results in this manuscript are deterministic evaluations of the variance formulas given fixed input parameters; no Monte Carlo simulation is used and no random seed is required.*

[33] **Source code, companion manuscripts, and the shared bibliography are located at documented repository paths.**

- The R package source is under R/.
- Companion manuscripts (Papers 1, 2, 4) are under analysis/report/{01,02,04}-*/.
- This manuscript source and the shared bibliography are under analysis/report/03-allocation/.

Data and code availability. *The R package source is at `R/`. The companion manuscripts (Papers 1, 2, 4) are at `analysis/report/{01,02,04}-*/`. The manuscript source and shared bibliography (`references.bib`, a symlink to `../../references.bib`) are at `analysis/report/03-allocation/`.*

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